

HTML Tables

HTML tables are used to organize and display data in rows and columns. They are suitable for structured information such as student results, class timetables, fee details, product lists, and attendance records.

Tables should be used for **tabular data**, not for designing the layout of an entire webpage.

1. Basic Table Elements

Element	Purpose
<table>	Creates the table
<tr>	Creates a table row
<th>	Creates a heading cell
<td>	Creates a data cell
<caption>	Provides a title for the table
<thead>	Groups the header rows
<tbody>	Groups the main data rows
<tfoot>	Groups the footer or summary rows

2. Basic Table Syntax

```
<table border="1">
  <tr>
    <th>Roll No.</th>
    <th>Name</th>
    <th>Marks</th>
  </tr>

  <tr>
    <td>101</td>
    <td>Aman</td>
    <td>85</td>
  </tr>

  <tr>
    <td>102</td>
    <td>Riya</td>
    <td>90</td>
  </tr>
</table>
```

Explanation

- <table> begins and ends the table.
- <tr> creates one horizontal row.
- <th> creates a heading cell, usually displayed in bold.
- <td> creates a normal data cell.
- border="1" displays the table border.

The `border` attribute is useful for basic HTML practice, but CSS is recommended for modern webpages.

3. Table Caption

The `<caption>` element provides a title for the table. It should be placed immediately after the opening `<table>` tag.

```
<table border="1">
  <caption>Student Result 2026</caption>

  <tr>
    <th>Name</th>
    <th>Marks</th>
  </tr>

  <tr>
    <td>Aman</td>
    <td>85</td>
  </tr>
</table>
```

4. Table Heading: `<th>`

The `<th>` element identifies a heading cell.

```
<tr>
  <th>Product</th>
  <th>Price</th>
  <th>Quantity</th>
</tr>
```

Browsers normally display table headings as bold and centre-aligned.

5. Table Data: `<td>`

The `<td>` element contains the actual table data.

```
<tr>
  <td>Laptop</td>
  <td>₹50,000</td>
  <td>2</td>
</tr>
```

Every `<td>` element should be placed inside a `<tr>` element.

6. Table Sections

A structured table can be divided into three logical parts.

`<thead>`

The `<thead>` element groups the table's heading rows.

`<tbody>`

The `<tbody>` element groups its primary data rows.

`<tfoot>`

The `<tfoot>` element groups summary or footer rows.

```
<table border="1">
  <caption>Product Sales</caption>

  <thead>
    <tr>
      <th>Product</th>
      <th>Quantity</th>
      <th>Amount</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>Laptop</td>
      <td>2</td>
      <td>₹1,00,000</td>
    </tr>

    <tr>
      <td>Mouse</td>
      <td>5</td>
      <td>₹2,500</td>
    </tr>
  </tbody>

  <tfoot>
    <tr>
      <td colspan="2">Total</td>
      <td>₹1,02,500</td>
    </tr>
  </tfoot>
</table>
```

These elements improve table organization and make the code easier to understand.

7. Merging Columns with `colspan`

The `colspan` attribute allows one cell to extend across multiple columns.

Syntax

```
<td colspan="number">Content</td>
```

Example

```

<table border="1">
  <tr>
    <th colspan="2">Student Details</th>
  </tr>

  <tr>
    <th>Name</th>
    <th>Course</th>
  </tr>

  <tr>
    <td>Aman</td>
    <td>CSE</td>
  </tr>
</table>

```

Here:

```
<th colspan="2">Student Details</th>
```

extends across two columns.

8. Merging Rows with `rowspan`

The `rowspan` attribute allows one cell to extend vertically across multiple rows.

Syntax

```
<td rowspan="number">Content</td>
```

Example

```

<table border="1">
  <tr>
    <th>Name</th>
    <th>Subject</th>
  </tr>

  <tr>
    <td rowspan="2">Aman</td>
    <td>HTML</td>
  </tr>

  <tr>
    <td>CSS</td>
  </tr>
</table>

```

The name Aman extends vertically across two rows.

9. Combining `rowspan` and `colspan`

Both attributes can be used in the same table.

```

<table border="1">
  <tr>
    <th rowspan="2">Roll No.</th>
    <th rowspan="2">Name</th>
    <th colspan="2">Marks</th>
  </tr>

  <tr>
    <th>Internal</th>
    <th>External</th>
  </tr>

  <tr>
    <td>101</td>
    <td>Aman</td>
    <td>40</td>
    <td>50</td>
  </tr>
</table>

```

Here:

- `rowspan="2"` makes Roll No. and Name extend across two header rows.
- `colspan="2"` groups Internal and External marks under Marks.

10. Styling Tables with CSS

CSS provides better control over table appearance than old HTML attributes.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Styled Table</title>

  <style>
    table {
      width: 70%;
      border-collapse: collapse;
    }

    table,
    th,
    td {
      border: 1px solid black;
    }

    th,
    td {
      padding: 10px;
      text-align: center;
    }

    th {
      background-color: lightblue;
    }

    caption {

```

```

        font-size: 22px;
        font-weight: bold;
        margin-bottom: 10px;
    }
</style>
</head>

<body>

    <table>
        <caption>Student Result</caption>

        <tr>
            <th>Roll No.</th>
            <th>Name</th>
            <th>Marks</th>
        </tr>

        <tr>
            <td>101</td>
            <td>Aman</td>
            <td>85</td>
        </tr>

        <tr>
            <td>102</td>
            <td>Riya</td>
            <td>90</td>
        </tr>
    </table>

</body>
</html>

```

Important CSS properties

Property	Purpose
border	Adds a border
border-collapse	Combines adjoining cell borders
padding	Adds space inside cells
text-align	Controls horizontal text alignment
background-color	Sets the background colour
width	Controls the table width
margin	Adds space outside the table
caption-side	Controls the caption's position

11. border-collapse

Without `border-collapse`, borders of adjacent cells may appear separately.

```

table {
    border-collapse: collapse;
}

```

It combines adjoining borders into a single border.

12. Cell Padding

Cell padding adds space between the cell content and its border.

```
th,  
td {  
    padding: 12px;  
}
```

13. Text Alignment

```
th {  
    text-align: center;  
}  
  
td {  
    text-align: left;  
}
```

14. Alternating Row Colours

Alternating row colours improve readability.

```
tbody tr:nth-child(even) {  
    background-color: #f2f2f2;  
}
```

The `:nth-child(even)` selector applies the style to even-numbered rows.

15. Row Hover Effect

```
tbody tr:hover {  
    background-color: lightyellow;  
}
```

The row's background changes when the user places the mouse pointer over it.

16. Complete Student Result Example

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">  
  
    <meta name="viewport"  
        content="width=device-width, initial-scale=1.0">  
  
    <title>Student Result</title>  
  
    <style>  
        table {  
            width: 80%;  
            margin: auto;  
        }  
    </style>  
</head>  
<body>  
    <table>  
        <tr>  
            <th>Sl. No.</th>  
            <th>Name</th>  
            <th>Marks</th>  
        </tr>  
        <tr>  
            <td>1</td>  
            <td>John Doe</td>  
            <td>85</td>  
        </tr>  
        <tr>  
            <td>2</td>  
            <td>Jane Smith</td>  
            <td>78</td>  
        </tr>  
        <tr>  
            <td>3</td>  
            <td>Mike Johnson</td>  
            <td>92</td>  
        </tr>  
    </table>  
</body>  
</html>
```

```

        border-collapse: collapse;
    }

    caption {
        font-size: 24px;
        font-weight: bold;
        margin-bottom: 10px;
    }

    th,
    td {
        border: 1px solid black;
        padding: 10px;
        text-align: center;
    }

    th {
        background-color: lightblue;
    }

    tbody tr:nth-child(even) {
        background-color: #f2f2f2;
    }

    tbody tr:hover {
        background-color: lightyellow;
    }

    tfoot {
        font-weight: bold;
        background-color: lightgreen;
    }
</style>
</head>

<body>

    <table>
        <caption>CSE Semester Result 2026</caption>

        <thead>
            <tr>
                <th rowspan="2">Roll No.</th>
                <th rowspan="2">Student Name</th>
                <th colspan="2">Marks Obtained</th>
                <th rowspan="2">Total</th>
                <th rowspan="2">Result</th>
            </tr>

            <tr>
                <th>Internal</th>
                <th>External</th>
            </tr>
        </thead>

        <tbody>
            <tr>
                <td>101</td>
                <td>Aman</td>
                <td>40</td>
                <td>50</td>
            </tr>
        </tbody>
    </table>

```

```

        <td>90</td>
        <td>Pass</td>
    </tr>

    <tr>
        <td>102</td>
        <td>Riya</td>
        <td>35</td>
        <td>45</td>
        <td>80</td>
        <td>Pass</td>
    </tr>

    <tr>
        <td>103</td>
        <td>Kabir</td>
        <td>20</td>
        <td>25</td>
        <td>45</td>
        <td>Fail</td>
    </tr>
</tbody>

<tfoot>
    <tr>
        <td colspan="3">Class Summary</td>
        <td colspan="3">Total Students: 3</td>
    </tr>
</tfoot>
</table>

</body>
</html>

```

Important Points

- Use `<table>` only for tabular data.
- Use `<tr>` to create rows.
- Use `<th>` for headings and `<td>` for ordinary data.
- Use `<caption>` to provide a table title.
- Use `colspan` to merge columns horizontally.
- Use `rowspan` to merge rows vertically.
- Keep the total number of cells consistent in each row after accounting for merged cells.
- Use `<thead>`, `<tbody>`, and `<tfoot>` to organize complex tables.
- Prefer CSS instead of outdated attributes for table styling.
- Provide clear headings so users can understand the data easily.

HTML Tables: Borders and Different Header Types

1. Table Border

The CSS `border` property adds a border around the table and its cells.

Syntax

```
border: width style color;
```

Example:

```
table,
th,
td {
    border: 2px solid black;
}
```

Here:

- `2px` is the border width.
- `solid` is the border style.
- `black` is the border colour.

2. Border Style

The `border-style` property specifies the appearance of the border.

```
table,
th,
td {
    border-width: 2px;
    border-style: solid;
    border-color: blue;
}
```

Common border styles

Border style	Appearance
<code>solid</code>	Continuous line
<code>dashed</code>	Dashed line
<code>dotted</code>	Dotted line
<code>double</code>	Double line
<code>groove</code>	Grooved three-dimensional border
<code>ridge</code>	Raised three-dimensional border
<code>inset</code>	Inset border
<code>outset</code>	Outset border
<code>none</code>	No border

Examples

```
border: 2px solid black;
border: 2px dashed blue;
border: 3px dotted red;
border: 4px double green;
```

3. Border Colour

The `border-color` property changes the border colour.

```
table,
th,
td {
    border: 2px solid;
    border-color: darkblue;
}
```

Colour values can be specified using:

```
border-color: red;
border-color: #2563eb;
border-color: rgb(0, 100, 200);
```

4. Border Radius

The `border-radius` property creates rounded corners.

```
table {
    border: 2px solid darkblue;
    border-radius: 10px;
}
```

For rounded corners to display correctly on a table, use:

```
table {
    border-collapse: separate;
    border-spacing: 0;
    overflow: hidden;
    border-radius: 12px;
}
```

Complete rounded-table example

```
<style>
    table {
        width: 70%;
        border: 3px solid darkblue;
        border-radius: 12px;
        border-spacing: 0;
        overflow: hidden;
    }

    th,
    td {
        border: 1px solid darkblue;
        padding: 10px;
        text-align: center;
    }
</style>
```

`border-radius` works more reliably with `border-collapse: separate` than with `border-collapse: collapse`.

Types of Table Headers

5. Horizontal Headers

Horizontal headers appear in the first row of a table.

```
<table border="1">
  <tr>
    <th>Roll No.</th>
    <th>Name</th>
    <th>Marks</th>
  </tr>

  <tr>
    <td>101</td>
    <td>Aman</td>
    <td>85</td>
  </tr>

  <tr>
    <td>102</td>
    <td>Riya</td>
    <td>90</td>
  </tr>
</table>
```

Expected structure:

Roll No.	Name	Marks
101	Aman	85
102	Riya	90

The `<th>` elements are placed horizontally in the first row.

6. Vertical Headers

Vertical headers appear in the first column. Each row begins with a `<th>` element.

```
<table border="1">
  <tr>
    <th>Name</th>
    <td>Aman</td>
  </tr>

  <tr>
```

```

        <th>Course</th>
        <td>CSE</td>
    </tr>

    <tr>
        <th>Semester</th>
        <td>Second</td>
    </tr>
</table>

```

Expected structure:

Name	Aman
Course	CSE
Semester	Second

Here, Name, Course, and Semester are vertical row headings.

7. Horizontal and Vertical Headers

A table can contain column headings in its first row and row headings in its first column.

```

<table border="1">
    <tr>
        <th>Student</th>
        <th>HTML</th>
        <th>CSS</th>
    </tr>

    <tr>
        <th>Aman</th>
        <td>85</td>
        <td>80</td>
    </tr>

    <tr>
        <th>Riya</th>
        <td>90</td>
        <td>88</td>
    </tr>
</table>

```

Expected structure:

Student	HTML	CSS
Aman	85	80
Riya	90	88

In this table:

- Student, HTML, and CSS are horizontal headers.
- Aman and Riya are vertical headers.

- Marks are placed in data cells.

8. Merged Header Using `colspan`

The `colspan` attribute merges two or more columns under one heading.

```
<table border="1">
  <tr>
    <th>Name</th>
    <th colspan="2">Marks</th>
  </tr>

  <tr>
    <th></th>
    <th>HTML</th>
    <th>CSS</th>
  </tr>

  <tr>
    <td>Aman</td>
    <td>85</td>
    <td>80</td>
  </tr>
</table>
```

Here:

```
<th colspan="2">Marks</th>
```

makes the heading Marks extend across the HTML and CSS columns.

9. Merged Header Using `rowspan`

The `rowspan` attribute merges a heading vertically across multiple rows.

```
<table border="1">
  <tr>
    <th rowspan="2">Student</th>
    <th colspan="2">Marks</th>
  </tr>

  <tr>
    <th>HTML</th>
    <th>CSS</th>
  </tr>

  <tr>
    <td>Aman</td>
    <td>85</td>
    <td>80</td>
  </tr>
```

</table>

Here:

- `rowspan="2"` makes Student cover two header rows.
- `colspan="2"` makes Marks cover two columns.

Complete Table with All Features

The following example includes:

- Border width
- Border style
- Border colour
- Border radius
- Horizontal headers
- Vertical headers
- Combined horizontal and vertical headers
- Merged headers using `rowspan` and `colspan`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>Student Result Table</title>

  <style>
    table {
      width: 75%;
      margin: 30px auto;
      border: 3px solid darkblue;
      border-radius: 12px;
      border-spacing: 0;
      overflow: hidden;
    }

    th,
    td {
      border: 1px solid #4682b4;
      padding: 12px;
      text-align: center;
    }

    th {
      background-color: #dbeafe;
      color: darkblue;
    }

    .main-header {
```

```

        background-color: darkblue;
        color: white;
    }

    .row-header {
        background-color: #e0f2fe;
        font-weight: bold;
    }

    tbody tr:nth-child(even) td {
        background-color: #f2f2f2;
    }

    tbody tr:hover td {
        background-color: lightyellow;
    }

    caption {
        margin-bottom: 10px;
        color: darkblue;
        font-size: 24px;
        font-weight: bold;
    }
</style>
</head>

<body>

    <table>
        <caption>Student Examination Result 2026</caption>

        <thead>
            <tr>
                <th rowspan="2" class="main-header">
                    Student
                </th>

                <th colspan="2" class="main-header">
                    Marks Obtained
                </th>

                <th rowspan="2" class="main-header">
                    Total
                </th>

                <th rowspan="2" class="main-header">
                    Result
                </th>
            </tr>

            <tr>
                <th>HTML</th>
                <th>CSS</th>
            </tr>
        </thead>

        <tbody>
            <tr>
                <th class="row-header">Aman</th>
                <td>85</td>
                <td>80</td>

```

```

        <td>165</td>
        <td>Pass</td>
    </tr>

    <tr>
        <th class="row-header">Riya</th>
        <td>90</td>
        <td>88</td>
        <td>178</td>
        <td>Pass</td>
    </tr>

    <tr>
        <th class="row-header">Kabir</th>
        <td>35</td>
        <td>30</td>
        <td>65</td>
        <td>Fail</td>
    </tr>
</tbody>

<tfoot>
    <tr>
        <th colspan="3" class="main-header">
            Total Students
        </th>

        <td colspan="2">3</td>
    </tr>
</tfoot>
</table>

</body>
</html>

```

Quick Summary

Feature	Code
Solid border	<code>border: 2px solid blue;</code>
Dashed border	<code>border: 2px dashed blue;</code>
Dotted border	<code>border: 2px dotted blue;</code>
Border colour	<code>border-color: blue;</code>
Rounded corners	<code>border-radius: 12px;</code>
Horizontal header	<code><tr><th>...</th></tr></code>
Vertical header	<code><tr><th>...</th><td>...</td></tr></code>
Merge columns	<code>colspan="2"</code>
Merge rows	<code>rowspan="2"</code>